The platform

Microsoft Fabric

C1 Fabric architecture

C1-L1 Tenant, capacity, workspace and item

The capacity decides who can view, how much runs at once and what it costs. Separate environments into workspaces, and give production its own capacity.

Fabric hierarchy	Tenant, capacity, workspace, item.
What F64 means	An F SKU with 64 capacity units. F SKUs run from F2 to F2048.
Free-licence viewers of Power BI content	Need an F64 or larger (or P) capacity. Smaller F SKUs need Pro, PPU or trial licences for viewers.
Pausing a capacity	Stops compute billing from then on, bills remaining smoothed and overage usage at once, and stops its items.

C1-L2 OneLake, one lake per tenant

One lake, one copy, many engines. Address data by GUID in anything automated.

```
https://onelake.dfs.fabric.microsoft.com/<workspace>/<item>.<itemtype>/<path>
abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.<itemtype>/<path>
```

How many OneLakes per tenant?	Exactly one. You cannot create or delete OneLakes.
OneLake table formats	Delta Parquet or Iceberg, open formats every Fabric engine reads.
OneLake path by name	abfss://<workspace>@onelake.dfs.fabric.microsoft.com/<item>.<itemtype>/Tables/<table>.
Why GUIDs in automation	They survive renames, avoid special-character problems in abfs URIs, and drop the item type.

C1-L3 Workspace roles, granted to groups

Roles go to groups, write access in production goes to automation, and people read.

Fabric workspace roles	Admin, Member, Contributor, Viewer.
Admin-only rights	Update or delete the workspace, manage admins, connect Git, create a workspace identity.
Viewer and lakehouse data	Reads with T-SQL (ReadData). Not through Spark or OneLake APIs (ReadAll starts at Contributor).
Who writes to production?	An automation identity as Contributor. People get Viewer, through groups.

C2 Lakehouse, Warehouse and SQL analytics endpoint

C2-L1 Lakehouse Tables and Files, and schemas

Raw files in Files, Delta tables in Tables, and schema-qualified names in shared code.

Lakehouse Tables vs Files	Tables: managed Delta tables, visible to SQL and Direct Lake. Files: raw and non-Delta data.

Non-Delta folder under Tables	Shown in the Unidentified area, not queryable.
Lakehouse schemas	On by default for new lakehouses. dbo is the fixed default schema.
Spark four-part name	workspace.lakehouse.schema.table. A bare name means dbo in the default lakehouse.

C2-L2 Warehouse T-SQL, and where it differs from SQL Server

Map the types, declare keys as NOT ENFORCED hints, and check uniqueness yourself after every load.

Primary keys in Fabric Warehouse	Only PRIMARY KEY NONCLUSTERED ... NOT ENFORCED. Duplicates are not blocked.
nvarchar, money, datetime in Fabric Warehouse	Not supported. Use varchar, decimal, datetime2.
MERGE and TRUNCATE in Fabric Warehouse	Both supported. MERGE is generally available.
Triggers in Fabric Warehouse	Not supported. Put the logic in the loading procedure.

C2-L3 Lakehouse, Warehouse or SQL analytics endpoint

The data and the logic are the same. The item type decides whether the procedure runs at all, and whether a failure leaves the tables consistent.

Lakehouse, Warehouse or endpoint: the deciding question	Who writes the data, and in which language? Spark writes to a lakehouse, T-SQL writes to a warehouse.
What can you do on the SQL analytics endpoint?	SELECT and create views, functions and procedures. No INSERT, UPDATE or DELETE on table data.
When do you need a Warehouse over a Lakehouse?	T-SQL writes, stored procedure loads, and multi-table transactions.
Read a lakehouse table from a warehouse in the same workspace	Use a three-part name: lakehouse.schema.table. No copy.
A new lakehouse table is missing from the endpoint	Metadata sync lag. Usually under a minute. Refresh or retry before dependent steps.

C3 Unifying datasets

C3-L1 OneLake shortcuts, and the delete that goes through

Shortcuts save copies. Treat every path under a shortcut as someone else's data.

What a shortcut is	A folder that points to data in OneLake or external storage, without copying it.
Delete the shortcut vs delete inside it	Deleting the shortcut keeps the data. Deleting files inside it deletes them in the target, if you can write there.

Shortcut as a table	Create it under Tables, pointing at Delta data.
External shortcut caching	OneLake caches read files for 1 to 28 days to cut egress.

C3-L2 Mirroring operational databases into OneLake

Mirroring replicates operational databases for free compute and generous storage, within per-source limits. Reconcile the table list after setup.

Shortcut or mirroring?	Shortcut for open-format data in storage. Mirroring for databases and catalogs, replicated when proprietary.
Mirroring cost	Replication compute is free. Storage is free up to 1 TB per CU. Queries use capacity.
Azure SQL Database mirroring limits	Up to 1,000 tables. Primary keys on datetime2(7), datetimeoffset(7) or time(7) block a table.
After setting up a mirror	Reconcile table lists and row counts with the source.

C3-L3 Mirroring Azure Databricks Unity Catalog into Fabric

Mirroring Unity Catalog copies no data and no permissions. Decide what is shareable, scope the connection, and rebuild access rules in Fabric.

What mirroring Unity Catalog moves	Only catalog metadata. Data is read in place through shortcuts.
Unity Catalog tables that do not mirror	Tables with row filters or column masks, federated, Delta Sharing and streaming tables, views and materialized views.
Unity Catalog permissions in Fabric	Not carried over. Rebuild access with Fabric's permission model.
Which identity reads mirrored data?	The connection's credential, for all queries. Scope it tightly.